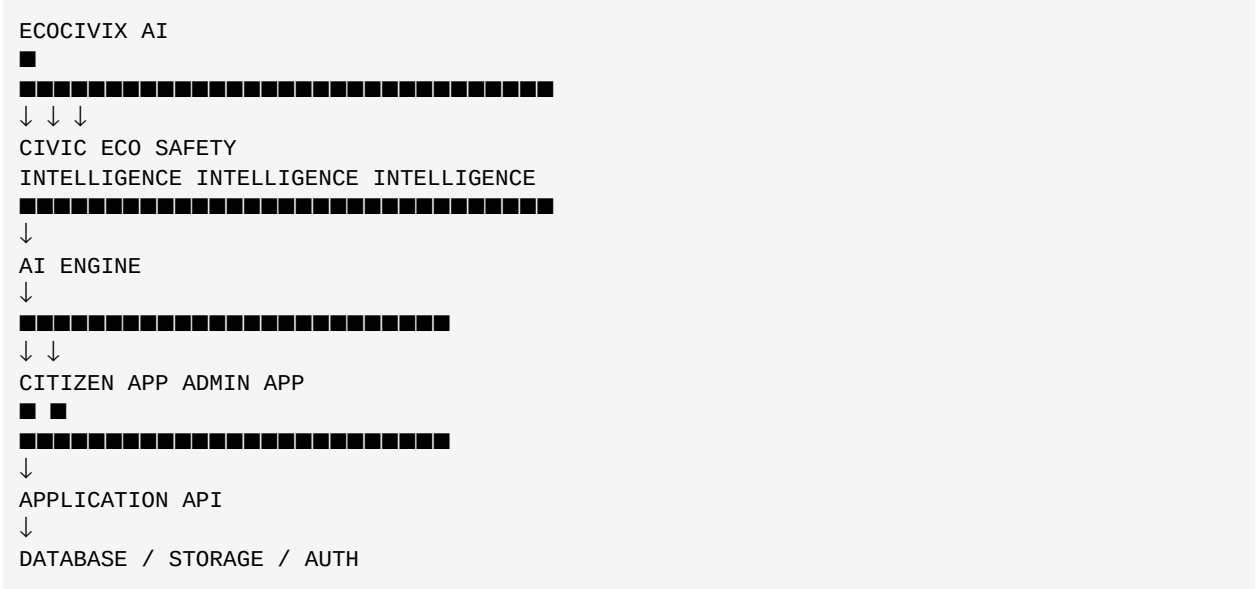


ECOCIVIX AI — ARCHITECTURE OVERVIEW

Pre-Hackathon Planning Document

1. High-Level Architecture



2. Preferred Technical Direction

- Frontend: React + Vite + JavaScript + CSS
- Backend: Node.js + Express + JavaScript
- Database: Supabase/PostgreSQL
- Authentication: Supabase Auth
- Storage: Supabase Storage where required
- AI: Gemini/free-tier provider where available
- Maps/location: suitable free/open solution where practical
- Deployment: free/low-cost services where available

3. AI Pipeline

Input → validation → issue understanding → classification → priority → environmental assessment → safety assessment → department/action → structured validation → application.

4. Data Flow

Citizen submits report → backend validates → AI analyzes → structured issue is persisted → admin reviews → status updates are persisted → citizen sees updates → aggregated data powers community intelligence.

5. Architecture Principles

- Real persistence
- Real authentication
- Server-side secrets
- Structured AI output
- Validated inputs
- Clear ownership
- Testable modules
- No fake final APIs